

BEHAVIOR

An animal's behavior encompasses all the things that it does as well as the way that it does them. Behavior ranges from simple actions, such as eating or keeping clean, to highly elaborate activities, such as hunting in a pack, courting a mate, or building a nest. In some animals, behavior is almost entirely predictable; in others, it develops with experience, so the more an animal does something, the more skilled it becomes. As with all aspects of animal biology, behavior is the product of evolution, which means that it gradually changes as time goes by. These changes enable species to react in the most effective way to the opportunities and dangers that they encounter in daily life.

Instinct and learning

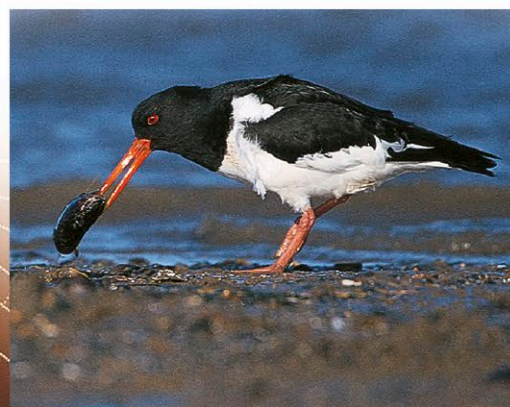
In simple animals, behavior is “hard wired”—governed by inherited instincts—which means that it consists of fixed sequences of actions prompted by triggers. For example, day-old birds instinctively beg for food when their parents appear at the nest. At this stage, they are usually blind: their behavior is triggered by noise and movement rather than by the sight of food itself.

Instinctive behavior may seem basic, but it can produce quite remarkable results. The structures that animals build—from nests to dams—are the results of inborn behavioral impulses. When beavers set out to make a dam, they do so without any knowledge of engineering principles. Yet the structure they make is shaped to withstand water pressure as if it had been scientifically designed.

Beavers do not have to think how to build, just as spiders do not need to work out how to weave webs. Even so, the results of instinctive behavior can change. As animals repeat certain tasks, such as making a nest, their performance often

improves. This is particularly important for some animals—such as male weaverbirds—which use their nest-building skills to attract a mate.

Apart from octopuses and their relatives, most invertebrates have narrow limits when it comes to learning. For vertebrates, on the other hand, learned behavior is often extremely important. Frogs and toads quickly learn to avoid animals that taste unpleasant, while mammals acquire a wide range of skills from their parents, including how to hunt. Among primates, individuals very occasionally “invent” new behavior, which is then copied by their neighbors. This copying process produces culture—patterns of behavior that are handed on down the generations. Culture is something that humans, as a species, have developed to a unique degree.



LEARNING TO FEED

Eurasian oystercatchers learn how to feed by watching their parents. Some birds hammer at shells to break them, while others stab at the shells' hinges to force them to open. Once a bird has learned one technique, it uses it for life.

INSTINCTIVE WEAVING

Spiders often build highly complex webs, but they always produce them to one particular design. As a result, it is often possible to identify a spider from its web alone. This garden spider will use its web for just one day; then it will eat it before starting on a replacement.

ANIMAL INTELLIGENCE

At one time, intelligence was thought to be rare in animals, with some exceptions such as dolphins, monkeys, and chimpanzees. Mammals include gifted communicators, but they are not alone in having problem-solving skills. Many other animals use simple tools, and some even shape them as well. Among the most impressive tool-makers are crows and their relatives. In captivity, one New Caledonian crow was seen bending a straight piece of wire, which it successfully turned into a hook for reaching food. It did this despite never having seen wire before—a remarkable example of insight that even a primate would find hard to match.



THE SIGNS OF INTELLIGENCE

Chimpanzees can solve problems, and they are able to learn sign language to communicate with humans. They can learn symbols for objects and actions, and they occasionally combine the symbols in ways that resemble spoken phrases.

Communication

For most animals, keeping in touch with their own kind is essential to their survival. Animals communicate with each other for a range of reasons, including finding food, attracting a mate, and bringing up their young. Different methods of communication have their own advantages and drawbacks. Body language—which includes facial expressions and physical displays—works well at close quarters but is ineffective at a distance and in habitats where dense vegetation gets in the way. In such cases, communication by sound is much more practical. Whales call to each other over immense distances, while some small animals produce remarkably loud sounds for their size. Treefrogs, cicadas, and mole crickets, for example, can often be heard at a distance of over 1 mile (2 km). Each species uses its own distinctive “call sign,” and many behave like ventriloquists, pitching their calls in a way that throws predators off their track.

Animals that are capable of producing light also use identifying call signs. These can consist of specific sequences of flashes or—in many deep-sea fishes—illuminated body patterns. But, like body language and sound, this form of communication works only when the signaller is actively signaling. Scent communication is quite

NIGHT LIGHT

Female glow worms use light to signal their presence to the males, which fly overhead. Signaling with light can be dangerous because it can attract predators as well as potential mates. If a glow worm senses danger, she quickly “switches off” her light.

